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CONCEPTIONS OF INSTINCT IN LANGUAGE AND COMMUNICATION: PRAGMATIST VS. RATIONALIST APPROACHES

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Abstract:

The concept of instinct has long played a central but controversial role in explanations of human cognition and behavior. In biology, psychology, and cognitive science, instinct is often invoked to explain capacities that appear early in development, recur widely across individuals, and emerge without explicit instruction. Yet critics have frequently argued that the concept is theoretically ambiguous and risks functioning as a placeholder explanation when underlying mechanisms remain unclear. This chapter reexamines the role of instinct in contemporary debates about language and cognition by contrasting two influential conceptions: Charles Sanders Peirce's pragmatist account of instinct as a form of abductive inference and Noam Chomsky's rationalist interpretation of instinct as innate cognitive structure underlying language.

After tracing the historical development of the instinct concept and reviewing major definitions used in modern research, the chapter surveys arguments both for and against instinctive explanations of cognition. Evidence from comparative biology, infant cognition, and evolutionary theory supports the

existence of biologically grounded predispositions, while critiques from developmental systems theory, anthropology, and philosophy of biology emphasize the importance of environmental interaction and cultural transmission in shaping cognitive development.

The central comparison focuses on Peirce's theory of instinctive abduction and Chomsky's theory of Universal Grammar. The difference between these two approaches to instinct in language and communication can be succinctly captured by the two fundamental rules that best captures the essence of each theory, The Basic Operation (Merge) for Chomsky and The Foundational Habit (Semeiosis) for Peirce, both explored below. Whereas Chomsky interprets instinct as structured cognitive architecture that constrains the acquisition of grammar, Peirce understands instinct primarily as an evolutionary tendency that guides hypothesis formation in inquiry. The chapter argues that Peirce's account provides a more flexible framework for integrating biological predispositions with learning and cultural evolution. It concludes by proposing that instinct is best interpreted as probabilistic cognitive bias rather than fixed innate structure, thereby offering a synthesis that accommodates insights from linguistics, philosophy, and cognitive science.

1. Introduction: Instinct as a Problem in Cognitive Science

The concept of instinct has occupied an unusually unstable place in the human sciences. Few terms have seemed at once so indispensable and so vulnerable to criticism. In biology, psychology, philosophy, and linguistics, “instinct” has repeatedly been invoked to explain patterns of behavior or cognition that appear early in development, recur widely within a species, and seem resistant to explicit instruction. Yet those same explanatory successes have often generated suspicion. When theorists say that an organism behaves “by instinct,” it is not always clear whether they are identifying a real mechanism, redescribing an observable regularity, or merely naming a phenomenon that they do not yet know how to explain. The concept is thus both durable and contested: durable because it answers to genuine empirical patterns, contested because its explanatory status remains uncertain.

This tension becomes especially acute in discussions of human cognition. In ordinary discourse, instinct often suggests something automatic, inflexible, and biologically fixed. In scientific discourse, however, the term has been used for a far wider range of phenomena[1]: species-typical action patterns in ethology (Blumberg 2016), inherited motivational dispositions in psychology (Hopwood, et. al. 2011), domain-specific learning constraints in cognitive science (Talevich, et. al. 2017), and even broad evolutionary predispositions that merely bias attention or inference (as Peirce proposed,

[1] Even the Oxford English Dictionary (OED) provides six definitions of "intuition," three of which it labels as "obsolete." One obsolete view has "intuition" as a synonym for contemplation or inspection. Another is "regarding as a motive for action". But the definitions of intuition still relevant to contemporary cognitive science and philosophy are from scholastic philosophy, modern philosophy - direct or immediate insight. Another definition, 5a in the OED, is common in modern philosophy and linguistics "... the immediate apprehension of an object by the mind without the intervention of any reasoning process."

see below). Such conceptual breadth has made the term useful, but it has also made it slippery. One of the central aims of this chapter is therefore not simply to defend or reject instinct, but to clarify what kinds of explanatory work the concept can legitimately perform with regard to language and communication.

The contemporary prominence of the concept owes much to the fact that several of the most important questions in cognitive science can be framed as disputes over instinct. Are there innate structures that shape perception (Dehn, et. al. 2025; Marr 1982)? Are there biologically prepared modes of social cognition (Jellema, et.al. 2024)? Are moral judgments partly instinctive (Hauser 2006)? Most dramatically, does language acquisition depend upon a specialized instinctive endowment (Pinker 1994)? In each case, the underlying issue concerns the relation between biological inheritance, learning, and the cultural environment. The more impressive the phenomenon appears to be—rapid, creative, robust, and cross-culturally recurrent—the stronger the temptation to appeal to instinctive structure. The more variable, developmentally plastic, and culturally dependent it appears, the stronger the motivation to resist such appeals.

Language is the central case because it lies at the intersection of all these pressures. Children acquire language rapidly, with relatively little explicit instruction, and do so in every known human society. The creative character of linguistic competence—our ability to understand and produce indefinitely many novel utterances—has often seemed to exceed what could plausibly be learned from finite and imperfect input. For this reason, language has become the most influential modern testing ground for claims about instinctive cognitive structure. The question is not merely whether language learning is biologically enabled; few would deny that. The sharper question is whether the mind contains richly specified grammatical or conceptual structures prior to experience (Carey 2004; Spelke 2022), or whether more modest biological biases, operating within powerful learning and cultural systems, are sufficient to explain acquisition.

Within this broader debate, two contrasting traditions are especially illuminating. One is associated with Noam Chomsky and the rationalist revival in modern linguistics that began in the late 1950s. Chomsky's generative program argues that the rapid and uniform acquisition of language points to an innate language faculty structured by Universal Grammar.[2] In this framework, instinct is understood primarily as internal cognitive architecture: a biologically given system that constrains the form of human grammars and narrows the hypothesis space available to children. The other tradition is associated with Charles Sanders Peirce, who was in fact the first American to use the term Universal Grammar (in 1865, in order to describe his own work on language; see Everett 2024). Unlike Chomsky, however, Peirce traced the tradition of universal grammar not back to 17th century Cartesianism, but back to the 13th century scholasticism via the work of the Franciscan monk Roger Bacon (1219-1292). Peirce did not develop a theory of language acquisition in the modern sense, yet he formulated a profound account of instinct as the guiding resonance and habit in abductive inference. For Peirce, instinct does not consist in detailed innate knowledge. Rather, it is an evolved tendency that helps organisms, and especially inquirers, form plausible hypotheses about the world. And it can also, given how the sense of the term has evolved over time, refer to what Everett (2016) refers to as "dark matter of the mind" (knowledge that is

[2] "A consideration of the character of the data actually available to the child suggests that the child must have a highly restrictive initial state of the language faculty." (Chomsky 1965: 58)

inferentially and usually subconsciously learned from an individual's enveloping cultural environment).

The contrast between these traditions is philosophically revealing. On the one hand the rationalistic theory treats instinct as structured endowment: an internal system that makes certain forms of cognition possible by fixing the range of acceptable representations. On the other hand, Peirce’s pragmatism (a theory he invented, not James nor Dewey, though the adapted Peirce's ideas in other directions) treats instinct as fallible guidance: a biologically rooted but corrigible tendency that directs inquiry for the language learner or the scientist. The first model emphasizes internal structure; the second emphasizes adaptive inferential orientation. Both approaches acknowledge that human cognition cannot be understood as the product of unconstrained learning from experience alone. But they differ fundamentally in how they conceive the form, scope, and epistemic role of innate tendencies.

The significance of this contrast extends beyond the history of ideas. Contemporary debates over usage-based linguistics, construction grammar, developmental systems theory, cultural evolution, and the poverty of stimulus (POS) argument can all be read, at least in part, as disputes over which conception of instinct is most defensible. Which of these approaches is correct (or might it be a combination of the two or another option entirely). The eventual answer matters not only for theories of language and communication, but for our broader conception of mind, knowledge, and human evolution.

This chapter argues that some conceptualization of instinct likely remains useful, but only if sharply distinguished from stronger and weaker versions of innateness. A central claim of the discussion is that Peirce's account of instinct as biases that guide abductive inference, as seen in what I will refer to below as The Foundational Habit, offers a more flexible, philosophically fruitful, and cross-species-relevant model than stronger conceptions of instinct as fixed cognitive structure. That does not mean that the Chomskyan tradition is wrong to insist on biological constraints in cognition. On the contrary, one of the lessons of modern research is that learning never occurs in a biologically neutral space. But it does suggest that the most defensible conception of the role of biology in language and cognition will leave room for fallibilism, developmental plasticity, and the shaping force of culture (Everett 2012, 2016, 2017).

The chapter proceeds in four stages. The first clarifies the historical development of the instinct concept and the major definitions employed in modern research. The second surveys the principal arguments for and against instinctive explanations, especially in relation to language and cognition. The third develops the contrast between Peirce's theory of "instinctive" abduction and Chomsky's rationalist linguistics, situating both within contemporary language-acquisition debates. The fourth proposes an alternative framework in which instinct is interpreted as probabilistic cognitive bias on inference rather than as inflexible structure. By approaching instinct through this comparative and interdisciplinary route, the chapter seeks not merely to adjudicate an old controversy, but to refine the conceptual tools with which that controversy is conducted.[3]

In some of my previous work on this topic (Everett 2012; Everett 2016; Everett 2024), I argued that instinct can, both in popular and scientific usage, see its meaning shift ambiguously between any subconscious knowledge and strictly phylogenetically

[3] Inference itself is found in all living species. The question of how that ability is innate, as it must partially be, is not addressed in this paper in any detail.

acquired habits, dispositions, or even in some of the literature, concepts, depending on the author.

1.2. Chomsky's Basic Operation and the Cartesian conception of Universal Grammar[4]

Avram Noam Chomsky's principal contributions to the understanding of instinct in the cognitive sciences include: (i) a reboot of Rationalism (Chomsky 1968); (ii) the Chomsky hierarchy of grammars (Chomsky 1956); and (iii) the adoption of the thirteenth century Scholastic term, Universal Grammar. However, Chomsky's concept of UG contrasts strongly with that of Bacon. Although for Bacon (and Peirce) UG was a matter of the logical constraints on signs and their arrangements, for Chomsky it is biological, what Chomsky 1965 referred to as "the initial state of the language faculty," i.e. *whatever* it is that is specific to human language at birth.[5] The "whatever" part of the understanding of UG at birth began a research program that has been going strong now for nearly seventy years.

The proposal of an initial state amounts to the very reasonable idea that no normal brain is a blank slate and that learning a language requires a biological basis. All learning requires a starting point. So nothing surprising about Chomsky's concern with an initial state.

[4] "If linear order is restricted to the mapping to the phonetic interface, then it gives no reasonto require the basic operation Merge to depart from the simplest form. As noted, many asymmetriecan be derived from configurations, from properties of lexical items, and from the independentlyneeded principles of labeling, so need not—hence should not—be specified by the elementaryoperation that forms the configuration. If that is universal, then the basic operation will be thesimplest possible one: unstructured Merge, forming a set." (Chomsky 2005, 15)

[5] As seen directly, some of us deny that there is any such thing.

The initial state by the mid 1960s was considered to be a set of rules, including both Phrase Structure (PS) Rules and Chomsky's innovation, Transformational (T) Rules, or just Transformations (a term that Chomsky adapted from his PhD advisor, Zellig Harris, at the University of Pennsylvania). To illustrate, rule (1) below is a PS rule and rule (2) is a T rule, to derive *Bill was seen by John* from *John saw Bill*:

(1) Phrase Structure Rule: S--> NP VP ("The symbol Sentence is rewritten/composed of a subject noun phrase (John) followed by a verb phrase (which itself contains the verb, saw, and the object noun phrase, Bill.")

(2) Transformation Rule: Active to Passive Rule

Rules of type (2) map the output of one PS rule like (1) onto another. So the in the pair of sentences just given, *John saw Bill* vs. *Bill was seen by John*, in order to capture the fact that these both mean the same thing, we say that *John saw Bill* is the "Deep Structure" and *Bill was seen by John* is the "Surface Structure" in the passive transformation. All of this has been subsequently abandoned, but it is what made Chomsky famous and it does still give the spirit of his research, which might be summarized as "What you see or hear in the syntax is not all there is in the syntax."[6],[7]

Both PS rules and T rules were therefore added to the research agenda of Chomskyan theory as part of the initial state of the newborn (though we also know now that learning begins in the womb and is guided by some vague innate biases that Peirce might have called *icons*). A full T rule for the passive derivation just discussed in Chomsky's early work (e.g. Chomsky 1955) is:

[6] Something Chomsky's teacher Harris would have denied and which over the years has been shown to be false, thus the "meaning preservation" property of T rules was abandoned by Chomsky.
[7] It should be remembered that when Chomsky talks of language, he is referring to the syntax, not the language as a whole.

(3) NP_1 - V - NP_2 --> NP_2 - Auxiliary Verb - V - en (by NP_1)

We read three as taking (1) above to output a structure in which what was the second NP becomes the first NP and the first NP becomes optional, preceded when it appears by the preposition *by* while the verb takes its passive form, indicated by the abstract suffix, -en.

In the early days of Chomskyan theory, this level of detail for the intial state was considered plausible and interesting. It was hypothesized that the passive, for example, should be more difficult for speakers and hearers to process than the active sentence, because the active sentences was the "kernel" sentence (Chomsky 1955) or Deep Structure (Chomsky 1965) and the passive was the *derived* (or Surface Structure) sentence. And for years, therefore, in both psycholinguistics and linguistics, this proposal was taken very seriously, that is, the idea that the greater the number of derivational steps required to derive a surface structure, the harder the sentence would be to process (an idea beginning with George Miller's work in the late 1950. Cf. also Fitch, Frederici, and Hagoort (2012) for the continued significance of the idea that became known as "derivational complexity"). Eventually, however, the failure to derive any significant results from the concept of derivational complexity, along with the dramatic proliferation of rules, transformations, and constraints on both PS rules and T rules led Chomsky and his followers to search for a simpler conception of the initial state, based on cross-linguistic and psycholinguistic data.

This led Chomsky to propose his now popular "Minimalist Program," to whittle linguistic theory down to "conceptual necessity." What emerged as conceptually necessary was, in particular, what Chomsky referred to as "The Basic Operation (TBO)," or Merge (intrinsically recursive), coming to base his entire theory on a form of set theory. As Chomsky describes Merge (Chomsky 2005, p6), "Merge takes two syntactic objects α and β and forms the set {α, β}."[8]

In Hauser, Chomsky, and Fitch (2002 1569) Merge is conflated (without comment) with recursion (since Merge is inherently recursive) to drive the following statement: "recursion ... is the only uniquely human component of the language faculty." Many researchers, myself included, take this to mean that UG has either been replaced by TBO or that it is unclear now what role UG plays (and this is still the case, thirty-one years into the Minimalist Program, in my interpretation of the intervening literature). That is, when the authors discussed "recursion," as subsequent replies to my challenges of this proposal made clear, what they meant was not recursion in a general mathematical or computational sense but the recursive set-theoretic operation of Merge. So after nearly seventy years of research on the intial state of the language faculty, the settled opinion of Chomsky and his followers seems to be that UG is TBO (Merge) and little else. There is a vast literature now on this, but this summarizes the findings so far.

In my own work (Everett 2008, 2012, 2016 and 2017), I have argued, to the contrary, that there is *no* initial state of any language faculty, because there is *no* linguistic faculty.[9] Humans learn based on semeiotic principles common to all species, but with a much larger brain.[10]

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[8] In 2007 I organized the first-ever conference on Recursion at Illinois State University, which drew researchers from many parts of the world and from many backgrounds, e.g. linguistics, mathematics, philosophy, psychology, and computer science. Those papers were edited and published as Van Der Hulst (2010).

[9] See https://daneverettbooks.com/research-scholarship/recursion/ for a list of my publications on recursion, and its implications for linguistic theory.

[10] And see Everett 2017 on brain evolution as well as many of the papers by the laboratory of Evelina Fedorenko at MIT's Brain and Cognitive Sciences Department, https://www.evlab.mit.edu.

The debate between me and others on UG has been going on for more than twenty-one years now. But rather than relitigating that particular and ongoing chapter of linguistic theory, I want to discuss an alternative to Chomsky's TBO and UG, in the form of what I will call (to be symmetrical) The Foundational Habit (TFH) of Peirce and Peirce's concepts of instinct and UG.

1.3. The Foundational Habit and Peirce's Semeiotic Conception of UG

The preceding sections have claimed that there are two influential yet fundamentally different approaches to the role of instinct in human cognition and language. We examined Chomsky's proposal for language (with potential applications to other areas of cognition that are not discussed in this chapter). Now let's look at Charles Sanders Peirce's pragmatist account via TFH, in which his version of instinct guides the formation of hypotheses through abductive reasoning, and Noam Chomsky's generative linguistics' account of instinct as biologically specified cognitive structure underlying grammar.[11] Although both thinkers acknowledge innate aspects of cognition, their theories differ sharply in how instinct operates, how knowledge develops, and how biological evolution relates to human intellectual capacities. Examining these differences clarifies the philosophical stakes of contemporary debates about language and cognition.

[11] For Peirce, human abductive inference or "instinct" was derived from the two human necessities, breeding and hunting. From the former some understanding of psychology emerges while from the latter some understanding of physics. These enter phylogentically so are genetically present in sapiens, as they are in many other species in one form or another.

In Peirce's semeiotic theory, one sign must be interpreted via another sign.[12] This operation, which I refer to as TFH, may be formalized simply as:

SIGN --> SIGN

This states that for any sign (icon, index, symbol, etc. see T.L. Short 2019 for an excellent introduction to Peirce's semeiotics; also footnote 10. See my own introduction to Peirce's semeiotics here: https://ling.auf.net/lingbuzz/008176 and Everett in progress), there must be another sign to serve as its interpretant.

According to TFH, all of inference, all of language, all of meaning, all of physics, and all of science more generally is partially determined (following closely Peirce's theory of semeiotic inference, Everett 2024 and Everett in progress) by offering one sign as an interpretant for another sign. (For example, if I see or hear the sign "bachelor" I can offer the (phrasal) sign "unmarried man" as its interpretant. If I see a footprint I might return the sign "bear" as its interpretant. If I see one numeral, say 2, I might return the (again phrasal) sign 1+1 as it interpretant. And so on.)[13]

This is a recursive interpretational procedure. If correct, this is really all there is to inference in any domain, i.e. returning one sign as the interpretant for another. And thus semeiotic inferentialism becomes the basis of all animal behavior, all of science and, ultimately all of nature (Everett in progress), connecting human language to animal communication systems. The primary difference between these two systems, according to Everett (2017) is that among all humans symbols are open-ended and productive

[12] The best-known signs in Peirce's system of sixty-six signs are *icons* (corresponding to an object in some way, e.g. resemblance); *indexes* (pointing to an object culturally or physically); and *symbols* (conventional meanings that are derived exclusively from culture.

[13] Peirce did allow for the possibility that semeiotic inferences could enter phylogenetically rather than ontologically. But I haven't really seen any need for the former. It remains an open research question.

systems whereas among other species, to the degree that symbols are found or proposed, they will be closed systems (though cf. Slobodchikoff (2009) for arguments that prairie dogs do have language). Other than symbols, non-humans rely more on indexes and icons, while Large Language Models arguably rely strictly on indexes (return one sign based on the statistical likelihood of it following its predecessor, a type of "pointing" or indexical sign). If Peirce is correct, instincts as such are not needed to be very precise, if needed at all, because TFH will hone in on correct usage, form, and meaning in language.

This section explains The Foundational Habit and its role in Peirce's conception of UG, as well as why and how it extends to all cognition, not merely language, and how it can be more effective than TBO in accounting for human language acquisition and variation, as well as providing an evolutionarily plausible theory of the continuum between the communication systems of human *and* nonhuman animals.

Peirce defines a habit as a general tendency to act in certain ways under certain conditions. Habits are not single actions but law-like dispositions that govern possible future behavior. Peirce places habit in his category of Thirdness, the realm of law, mediation, and generality (CP 5.487; CP 1.409)[14]

Peirce also famously identifies belief with habit. In "The Fixation of Belief," he writes that belief is that upon which one is prepared to act, that is, a habit of action (CP 5.398–5.400). So a habit is a generalization. Thus belief is not merely intellectual assent but a disposition guiding conduct. Moreover, in Peirce's theory of inquiry, doubt produces inquiry, and inquiry aims to settle belief. The resolution of inquiry results in the

[14] Peirce's phenomenology (which he called Phaneroscopy) has three components - firstness (vague perception - such as unfamiliar sound), secondness (opposition to another unit and recognize of the individuality of the unit in focus, such as a phonetic segment), and thirdness (lawlike generalizations, e.g. as the role of a sound in a phonological analysis or theory).

formation of stable habits. Thus knowledge is best understood not as static propositions but as settled habits of action (CP 5.416; CP 5.417). In Peirce's semeiotic theory, the meaning of a sign lies in its interpretant. The ultimate interpretant is a habit of returning one sign for another —a general disposition to respond in certain ways. This connects meaning directly to habitual conduct (CP 5.476; CP 5.491) And this habit of using one sign as the interpretant of another is, again, TFH.

In fact, Peirce extends TFH, returning one sign for another, beyond human psychology into cosmology. He argues that the laws of nature themselves are habits that have developed over time. This doctrine of "habit-taking" presents the universe as evolving from chance to law (CP 6.101; CP 6.262)

TFH applies also and especially to learning. This is so because for Peirce, learning consists in the acquisition and modification of habits. Experience and repetition shape habits, while abduction introduces new possible habits. Intelligence thus is partially the capacity to form and revise habits (CP 5.171; CP 7.515)

In this system habits are subject to self-control and normative evaluation. Logical norms govern habits of reasoning, ethical norms govern habits of action, and aesthetic norms guide ideals. Thus habit is central to Peirce's normative sciences (CP 1.573; CP 5.533).

Before moving on, I want to conclude this section by discussing how TFH predicts the operation of Large Language Models. An LLM also is based on inferring along the following lines: From position n we can infer position n+1. But this is simply a variation of TFH! And as an added bonus (if the reader thinks of the problem as serious, as I do) this also helps us to better understand John Searle's famous Chinese room thought experiment. In this experiment, a "translator" is placed in a room. In front of this person are two openings. Into one opening comes Chinese characters. The task of the "translator" is to follow an instruction sheet and match each character coming in one opening with an English word then being pushed out the other opening according to the instructions. To the person outside the box it seems that inside of the box, our translator, understands Chinese and English. But in fact he or she need not understand either. This is so because, in Peirce's terms they are simply matching one sign (the Chinese character) up with another. That is, the Chinese character is an index of an English word. Indexes require no understanding of meaning. Only matching. So of course there is no understanding and this is predicted by Peirce's TFH because in this case the first sign is an index and the second is the object: SIGN$_{\text{INDEX}}$ --> SIGN, as opposed to SIGN$_{\text{SYMBOL}}$ --> SIGN$_{\text{SYMBOL}}$ (which is the inference based on meaning). To get understanding the interpreter would need to understand the Chinese character, i.e. interpret it as a symbol (SIGN$_{\text{SYMBOL}}$ --> SIGN). But since meaning was ex-hypothesi not programmed in, the machine can do the task just fine based on indexical inference rather than symbolic inference (both of which are subsets of TFH). The same is the case for LLMs as they currently are designed. There is the appearance of understanding without understanding. Peirce's semeiotics predicts this (Chomsky has nothing to say about this). But, even more importantly, Peirce's semeiotics explains why LLMs and Chinese rooms can do so well without understanding meanting - because they follow TFH at a different level of semeiosis (secondness - indexes vs. thirdness - symbols).

2. Historical Development of the Concept of Instinct

The modern concept of instinct, independently of Chomsky or Peirce, emerged from a long and uneven history in which questions about natural impulse, habit, rationality, and biological organization were repeatedly reformulated in new vocabularies. Although the term itself acquired its modern scientific weight only in the nineteenth century (in works e.g. Darwin 1859), the underlying problem is much older. Classical philosophers already confronted the contrast between behavior that appears guided by reason and behavior that appears to arise from nature itself. Aristotle, for example, distinguished the rational capacities of human beings from the more limited perceptual and appetitive powers of animals, while nevertheless recognizing that animals display patterned and purposive forms of action. Such action was not random: it exhibited order, regularity, and species-specific form. What later thinkers would call instinct was thus already visible as the problem of how non-rational creatures nonetheless behave in ways suited to their ends.

Stoic thought deepened this issue through the notion of natural impulse or oikeiōsis, according to which living beings are directed toward self-preservation and the maintenance of their proper form. In this framework, purposive behavior need not depend upon reflective deliberation. Organisms can be oriented toward what is fitting for them by virtue of their nature. Although ancient accounts lack the genetic and evolutionary assumptions of modern biology, they established a durable intuition: action can be adaptive, organized, and reliable without being the result of explicit reasoning.[15]

Early modern philosophy transformed the debate by tying it to the rise of mechanistic science. René Descartes offered one of the most influential, and

[15] Inference is the core of reason, but it need not be explicit, conscious, deliberative, or slow.

controversial, formulations when he described animals as automata (Descartes 1637, pV). In Cartesian thought, the apparently purposive behavior of animals could be explained through the organization of bodily machinery rather than through rational thought. The language of instinct increasingly became associated with mechanisms that produced reliable responses without conscious reflection. Yet the Cartesian framework also sharpened a contrast that would matter for later discussions: if animals are governed by mechanism while humans possess reason, where exactly is the boundary between mechanical-instinctive and rational action?

Empiricist philosophers, especially John Locke, complicated matters further by criticizing innate ideas. Locke’s attack was directed primarily at epistemology, but it shaped later suspicion toward claims of innateness more broadly. If knowledge is not stamped upon the mind at birth, then appeals to instinct must be framed carefully. The result was a lasting division between traditions that saw inherited structure as essential to cognition and those that worried such claims smuggled unexplained capacities into theory under the guise of natural endowment.

The nineteenth century decisively recast the discussion through evolutionary theory. Charles Darwin’s treatment of instinct was especially important because it naturalized the phenomenon without reducing it to mere mechanism in the Cartesian sense. In Darwin’s hands, instincts became inherited behavioral tendencies shaped by natural selection. The analogy to anatomy was explicit: just as bodily organs evolve because they contribute to fitness, so too do behavioral dispositions. Darwin’s work on instinct, including his discussions of social insects, migration, and emotional expression, gave the concept a central place in biology. Instinct was no longer merely a philosophical category marking the boundary between reason and impulse; it became an evolutionary explanandum and an adaptive achievement.

This Darwinian turn had two lasting consequences. First, it encouraged the search for species-typical behavioral patterns that could be explained historically in terms of selection. Second, it made it possible to ask whether human cognition itself might contain evolved instinctive dispositions. If instincts are adaptive inheritances, then there is no principled reason why the human mind should be exempt from such evolutionary shaping. The modern science of instinct begins here (cf. Wilson (1975); Buss (2024))

In early psychology, the concept expanded dramatically. William James famously argued that human beings possess many instincts, not fewer than animals but more, because their greater nervous complexity supports a wider range of inherited tendencies (James 1887).[16] Thus, curiosity, fear, sympathy, imitation, and social responsiveness could all be discussed as instinctive. William McDougall (McDougall, Shand, and Stout 1914) further systematized this approach by tying instincts to specific emotional and motivational systems. In this period, instinct served as a bridge concept linking biology, affect, motivation, and behavior. It also became a way of resisting overly intellectualist conceptions of mind by emphasizing that much human conduct begins from inherited tendencies rather than reflective calculation.

The rise of behaviorism in the early twentieth century brought a forceful reaction. John B. Watson (1914), and later B. F. Skinner (1957), regarded instinctive explanation with suspicion because it often seemed to halt inquiry rather than advance it. Why say that a behavior occurs "by instinct" if one can instead study the environmental

[16] In Everett (2016) I argue that James got it exactly wrong. Humans evolved, in my analysis, for *fewer* instincts and more freedom cognitively than other animals.

contingencies that shape it? Behaviorists did not deny that organisms differ biologically, but they argued that scientific explanation should prioritize observable relations between stimuli, responses, and reinforcement histories. In this climate, instinct acquired a somewhat disreputable status in some sectors of psychology, appearing too vague, too mentalistic, or too essentialist to count as a serious scientific construct.

Yet instinct never disappeared. Mid-twentieth-century ethology, especially in the work of Konrad Lorenz (1965) and Nikolaas Tinbergen (1951), restored legitimacy to instinct by studying it in a far more disciplined way. Ethologists focused on species-typical action patterns, sign stimuli, and fixed action sequences, often under naturalistic conditions. Their work demonstrated that certain complex behaviors could emerge reliably without the sort of learning histories emphasized by behaviorism. At the same time, ethology refined the concept: instinctive behaviors were no longer treated as mysterious inner drives, but as organized behavioral systems susceptible to empirical description.

The later twentieth century complicated the picture yet again. Cognitive science reintroduced innateness in new forms, especially through Chomsky's (1959) critique of behaviorist accounts of language learning. Developmental biology, along with anthropology, however, simultaneously challenged simplistic nature–nurture dichotomies by showing how developmental outcomes arise through complex interactions among genes, environments, and organismic activity. Thus the history of instinct is not a simple linear progression from superstition to science. It is a sequence of redefinitions, each shaped by changing models of life, mind, and explanation. As Peirce might put it, the study of instinct can only asymptotically approach Truth through dialogue.

This history matters for the present argument because it reveals that "instinct" has never had a single stable meaning. Over time, it has designated natural impulse, automatic mechanism, inherited behavior, motivational system, domain-specific cognitive structure, and adaptive bias. Any contemporary use of the term must therefore specify which historical lineage it inherits and which explanatory burden it is meant to bear. The comparison between Peirce and Chomsky is especially revealing precisely because they occupy different points in this historical field: Peirce inherits the evolutionary and logical problem of how organisms form successful hypotheses, while Chomsky inherits the rationalist and cognitive problem of how rich mental structure can arise in development.

3. Major Definitions of Instinct in Modern Research

Again, debates about instinct often generate confusion because the term is used to refer to several quite different explanatory concepts. Contemporary researchers do not simply disagree about whether instinct exists; they often mean different things by the word itself. A careful taxonomy of definitions is therefore necessary before the concept can be evaluated more generally, apart from Chomskyan or Peircean theory. In modern research, at least four broad families of usage recur: ethological, evolutionary-biological, psychological, and cognitive-scientific. Each captures something important, but none can simply be assumed to exhaust the phenomenon.

The ethological definition is perhaps the most concrete. On this view, instincts are species-typical behavioral programs or action patterns that develop reliably and are elicited under characteristic conditions (Bond, Barlow, and Rogers 1985; Ronacher

2019). The emphasis falls on observable regularity: what members of a species tend to do, how stereotyped the behavior is, and what stimuli release it. Classic examples include courtship displays, nest building, migration, or fixed action patterns studied by Lorenz and Tinbergen. The strength of this definition is that it anchors the concept in visible, comparative phenomena. Its weakness is that it may underdescribe the underlying mechanisms and can be difficult to apply to flexible or cognitively mediated behavior.

A second, somewhat broader, definition comes from evolutionary biology. Here instinct is understood as a genetically canalized behavioral tendency shaped by natural selection (cf. Blumberg 2016 for a more nuanced perspective, however). The central idea is not merely that a behavior is common, but that it reflects inherited developmental organization whose persistence has adaptive significance. This definition usefully situates instinct within a historical and population-level framework (cf. Loeb 2025).[17] It explains why certain patterns recur and why they should be expected to fit recurrent ecological problems. Yet it raises a familiar difficulty: genes do not code for fully formed behaviors in any simple one-to-one way. Once the gene–development–behavior relation becomes more complicated, the definition risks either oversimplification or metaphor.

[17] In Everett 2016 and 2017, I take a much more skeptical view of instinct: "... humans have evolved away from cognitive rigidity (a by-product of instincts), to cognitive flexibility and learning based on local cultural and even environmental constraints. Under these assumptions, any grammatical similarities found across the world's languages would not be because grammar is innate. Rather they would indicate either functional pressures for effective communication that go beyond culture to simple efficiency in information transfer. An example of a functional pressure is the fact that in most languages, prepositions with less semantic content are shorter than prepositions with more contact, as in the contrast between, say, "to" or "at" vs. "about" or "beyond." An example of efficiency in information transfer is seen in the fact that less frequent words are more predictable in their shape than those speakers use more frequently. So the verb "bequeath," as a less-frequent verb, has a simple conjugation: "I bequeath," "you bequeath," "she bequeaths," "we bequeath," and "everyone bequeaths (this general principle is known as "Zipf's Law"). But the common verb "to be," is "I am," "you are," "he is," "we are," and "they are." Both functional pressure and information transfer requirements optimize language for better communication." I will avoid more detailed arguments here.

Psychological definitions often shift attention from overt behavior to dispositional organization. On this view, instincts are innate motivational, affective, or cognitive dispositions that orient organisms toward certain classes of objects, responses, or goals. William James and William McDougall are paradigmatic figures here. The advantage of the psychological definition is that it can explain flexible behavior better than narrow ethological accounts. A creature may have an instinctive fear response, attraction, or social tendency without manifesting it in exactly the same way on every occasion. But this flexibility also produces ambiguity. The more an instinct looks like a broad disposition rather than a fixed behavior, the harder it becomes to distinguish it from temperament, drive, bias, or learned habit (which is all it is for Peirce throughout his writings, whether learn phylogenetically or ontogenetically).

A fourth family of definitions has been especially influential in cognitive science. Here instinct is redescribed as domain-specific computational architecture or innate representational structure. The most important modern example is the claim that the mind contains specialized systems (cf. Fodor 1980 for example), perhaps for language, face recognition, or social reasoning—that constrain what can be learned and how. This approach has the virtue of integrating instinct with formal models of cognition. It moves beyond visible behavior to hypothesized internal organization. But it also changes the meaning of instinct substantially. What is "instinctive" in this usage may consist not in overt action or even broad disposition, but in pre-specified computational principles (Marr 1982). The concept thus becomes more abstract and more controversial.

A great deal of debate arises because these definitions are often conflated. A researcher may cite stereotyped animal behavior in support of instinct and then move, too quickly, to the conclusion that humans possess richly specified cognitive modules. But the inferential step is not straightforward. Evidence that some behaviors are biologically canalized does not by itself show that complex symbolic systems are innately structured in the same way. Conversely, criticism of simplistic gene-for-behavior models does not by itself undermine the possibility of weaker innate biases. Disputes over instinct are therefore often disputes about the level of analysis at which the concept is applied.

Modern literature also introduces a number of more cautious reformulations. Some scholars prefer the language of "preparedness," "constraints," "biases," or "priors" rather than instinct. These alternatives are not merely terminological. They often signal a weaker claim: instead of saying that a behavior or form of knowledge is instinctive, one says that organisms are predisposed to acquire or perform it under certain conditions. This vocabulary is especially attractive in developmental and probabilistic frameworks because it avoids implying that behavior is fixed independently of environment. The organism begins with structured tendencies, but those tendencies interact with experience in complex ways.

Several conceptual problems cut across all these definitions. One is circularity. If instinct is identified merely by the appearance of spontaneity, early emergence, or species-typicality, then calling a behavior instinctive may amount to little more than restating the phenomenon. Another is the gene–behavior mapping problem. Behavioral traits are rarely traceable to discrete genetic causes in the way folk discourse about instinct sometimes suggests. A third is developmental plasticity. Many behaviors that appear innate depend upon environmental input, social context, or self-organizing developmental processes. The more plastic the trait, the more difficult it becomes to classify it cleanly as instinctive or learned.

These problems are not reasons to abandon the term altogether, but they do require methodological discipline. To say that a trait is instinctive should ideally specify at least four things: what aspect of the trait is inherited or developmentally biased, what mechanisms are proposed to underlie it, and how those mechanisms interact with environment and experience, and, last but by no means least, how they evolved by natural selection (Tinbergen 1951, among others). Without such specification, the word “instinct” risks doing rhetorical work in place of explanatory work.

These definitional distinction spaces are crucial because, again, Peirce and Chomsky occupy very different positions within them. Peirce’s view most closely resembles a theory of evolved inferential disposition or bias. Instinct, for him, is not a stock of built-in propositions or rules, but is largely a tendency of mind to generate plausible hypotheses via inference (awareness is not required for this to occur, according to Peirce).[18] For humans this can be science or, for both humans and non-humans, escaping predators, hunting, etc. Chomsky’s view, by contrast, fits most naturally within the cognitive-scientific conception of instinct as internal structure—an innate architecture that constrains what grammars can be acquired. The rest of this chapter explores the consequences of that difference. But before that contrast can be fully drawn, however, it is necessary to assess the broader arguments for and against instinctive explanations as such.

[18] Therefore non-conscious entities like thermostats or decidious trees during abcission can be said to infer the appropriate habit application from their environment, i.e. adjusting the temperature, shedding leaves, etc.

4. Arguments for Instinctive Explanations

Despite the substantial criticisms discussed in the previous section, many scholars continue to argue that instinctive or innate factors play an important role in explaining behavior and cognition. These arguments arise from several domains of research, including comparative biology, developmental psychology, evolutionary theory, and linguistics. Proponents of instinct-based explanations contend that certain patterns of behavior and cognition are difficult to account for solely through learning and environmental influence. Instead, they argue that biological inheritance provides structured predispositions that guide behavior and learning from the earliest stages of development.

4.1 Evidence from Animal Behavior

One of the strongest sources of evidence for instinctive behavior comes from comparative studies of animals. Ethologists such as Konrad Lorenz and Nikolaas Tinbergen documented numerous behavioral patterns that appear to develop reliably across members of a species without requiring extensive learning.

Classic examples include bird migration, nest building, mating displays, and fixed action patterns such as egg-retrieval behavior in geese. These behaviors often emerge in individuals that have had limited opportunities to observe others performing the same actions. In many cases, the behaviors appear to be triggered by specific environmental stimuli and executed in highly stereotyped ways.

For example, Lorenz's (1935) research on imprinting demonstrated that certain young birds develop strong attachments to the first moving object they encounter after hatching. This process occurs within a limited developmental window and appears to operate according to species-specific biological mechanisms (but cf. Blumberg (2016) for alternative perspectives on such proposals).

Such findings suggest that biological inheritance can produce complex behavioral patterns that unfold reliably during development. Although learning may refine these behaviors, their basic structure appears to be guided by inherited predispositions.

4.2 Early Human Cognitive Capacities

Evidence for innate influences on cognition also comes from studies of human infants. Developmental psychologists have identified a number of perceptual and cognitive capacities that appear to emerge early in life, often before extensive learning has occurred (but cf. Everett 2012).

For instance, infants show preferences for human faces shortly after (and even before (Ronga, et. al. 2025) birth and display sensitivity to certain visual and auditory patterns that are important for social interaction. Research on early numerical cognition (Cantrell and Smith 2013) suggests that infants possess rudimentary abilities to distinguish between small quantities. Similarly, as seen earlier, studies of language perception have demonstrated that infants can discriminate among phonetic contrasts from many different languages during the first months of life (Sundara et. al. 2018), though this is perhaps most economically understood as co-evolution of the ears and mouth (i.e. of the auditory and vocal apparatuses).

These early cognitive capacities have sometimes been interpreted as evidence for domain-specific learning mechanisms shaped by evolution. If infants begin life with structured perceptual and cognitive biases, these predispositions may guide the acquisition of more complex forms of knowledge.

Importantly, such findings do not imply that complex cognitive abilities are fully specified at birth. Rather, they suggest that development may begin with certain biases or sensitivities that shape how individuals interact with their environments.

4.3 Evolutionary Explanations of Cognitive Biases

Evolutionary theory provides another argument in favor of instinctive influences on cognition. Natural selection operates by favoring traits that enhance survival and reproduction within particular ecological contexts. Over evolutionary time, organisms may therefore develop specialized behavioral tendencies that help them navigate recurring environmental challenges.

In humans, evolutionary psychologists have argued that certain cognitive tendencies may reflect adaptations to problems faced by ancestral populations. Examples often discussed include preferences related to food selection, fear responses to potentially dangerous animals, and social cognition related to cooperation and competition. Pinker and Bloom (1990) offer a moderately plausible scenario of the evolution of such adaptations with regard to language.

Although the specific claims of evolutionary psychology remain debated, the general idea that evolution shapes behavioral predispositions is widely accepted in biology. Cognitive systems are themselves products of evolutionary history, and it would therefore be surprising if they did not incorporate biases that reflect adaptive pressures.

From this perspective, instinctive tendencies may function as starting points that guide learning rather than as rigid programs determining behavior. Biological predispositions can influence what individuals attend to, how they interpret information, and which patterns they learn most easily. Hard to imagine dissent from such a statement no matter the theoretical vantage point.

4.4 Biological Specialization and Cognitive Architecture

Another line of reasoning supporting instinctive influences on cognition arises from research on biological specialization. Many biological systems exhibit highly specialized structures adapted for particular functions. For example, the visual system contains neural mechanisms specialized for processing visual information, while the auditory system is specialized for detecting sound (Swanson 2011).[19]

If biological systems often evolve specialized structures for specific tasks, it is plausible that cognitive systems may also exhibit forms of specialization. Some researchers therefore argue that the human brain may contain domain-specific mechanisms that support particular cognitive abilities, including language.

Evidence from neuroscience (Fedorenko, et. al. 2024) has identified regions of the brain that play important roles in language processing. Although these regions do not operate independently of broader cognitive systems, their involvement suggests that language may depend in some way on partially specialized neural architecture, though Fedorenko is careful to avoid calling language an instinct or labeling grammar as innate.

From this perspective, instinctive influences on cognition may reflect biological constraints embedded within neural systems. These constraints might then shape how cognitive processes unfold without determining behavior in a rigid or deterministic manner.

4.5 Instinct as Developmental Bias

[19] But there are no known comparable structures for language more generally. On the other hand, Fedorenko's work on the language network suggests prespecified cellular structure for semeiotics - my interpretation, not necessarily hers. Numerous papers by her and her team along these lines are available here: https://www.evlab.mit.edu

Taken together, these arguments suggest that biological inheritance can influence behavior and cognition in meaningful ways. However, many contemporary researchers interpret instinct more modestly than earlier theorists did (Salazar-Ciudad 2021, among others).

Under this interpretation, organisms begin life with certain predispositions that shape how they interact with their environments. These predispositions influence attention, perception, and learning, thereby guiding the development of more complex behavioral patterns.

In the context of human cognition, such developmental biases may help explain why certain forms of learning occur reliably across individuals. For example, children acquire language rapidly across cultures, suggesting that the human cognitive system is particularly well suited - in some way - to learning linguistic patterns (but cf. Everett 2012, 2016).

Importantly, developmental biases do not eliminate the role of learning or culture. Instead, they interact with environmental input to produce cognitive outcomes. Instinctive predispositions may guide the direction of learning while leaving considerable flexibility for variation across individuals and cultures.

4.6 Relevance for Theories of Language and Cognition

Contemporary research increasingly recognizes that instinctive influences operate within complex developmental contexts. Biological predispositions interact with environmental input, cultural transmission, and learning processes to produce behavioral outcomes.

Understanding the strengths and limitations of instinct-based explanations therefore remains central to contemporary debates about the nature of human cognition and language.

5 Major Critiques of the Instinct Concept

Although the concept of instinct has played a central role in biology, psychology, and cognitive science, it has also been the subject of sustained criticism (see, inter alia, Blumberg 2016; Buller 2006). Throughout the twentieth and twenty-first centuries, scholars have raised conceptual, empirical, and methodological objections to instinct-based explanations of behavior and cognition. These critiques do not necessarily deny that biological inheritance influences behavior, but they challenge the idea that complex cognitive capacities can be adequately explained by invoking instinct alone. The criticisms fall into several broad categories, including conceptual problems with the term itself, developmental systems critiques, anthropological evidence of cultural variation, challenges from evolutionary psychology debates, and increasing recognition of complex gene–environment interactions in development.

5.1 Conceptual Problems with the Term "Instinct"

Early critics noted that instinct frequently functions as a placeholder explanation rather than a genuine causal account. When a behavior appears early in development or is widely shared across individuals, it is sometimes labeled instinctive without identifying the specific mechanisms responsible for producing it. In such cases, the appeal to instinct risks becoming circular: a behavior is explained by instinct simply because it appears instinctive.

This conceptual problem was already recognized by several early psychologists. Even William James, in spite of his proposal that humans possess numerous instincts, acknowledged that the concept could easily become overextended. Later critics argued that without clear criteria for identifying instinctive behavior, the term provides little explanatory power.

Contemporary discussions in philosophy of biology and cognitive science have therefore emphasized the need to distinguish carefully between different kinds of innate tendencies. Some scholars prefer to replace the term instinct with more precise descriptions of developmental constraints, learning biases, or evolved cognitive mechanisms.

5.2 Developmental Systems Theory

Another influential critique comes from developmental systems theory, which emphasizes the complex interactions between genetic, environmental, and developmental factors in shaping behavior, especially with regard to communication and signalling. Researchers such as Susan Oyama (2000), Gilbert Gottlieb (2007), and Paul Griffiths (2013) have argued that behavior cannot be understood simply as the expression of genetic programs or instinctive mechanisms.

According to developmental systems theory, traits emerge from dynamic developmental processes involving multiple interacting influences. Genes, cellular processes, environmental conditions, and social interactions all contribute to the development of behavior. As a result, it is often misleading to attribute behavioral traits solely to innate instincts.

Gilbert Gottlieb's work on probabilistic epigenesis illustrates this perspective. Gottlieb demonstrated that even behaviors traditionally regarded as instinctive—such as species-specific vocalizations in birds—can depend on subtle interactions between genetic predispositions and environmental experience. Developmental outcomes therefore reflect probabilistic processes rather than rigid genetic programs.

This perspective challenges traditional interpretations of instinct by emphasizing that behavioral traits cannot be separated from the developmental systems that produce them. Rather than asking whether a behavior is innate or learned, developmental systems theorists focus on the network of interactions that give rise to behavioral patterns. This would have been congenial to Peirce's views.

5.3 Cultural Variation and Anthropological Evidence

Anthropological research has also raised important questions about instinct-based explanations of human cognition and behavior. Human societies display enormous variation in cultural practices, social norms, and linguistic structures. This diversity suggests that many aspects of human behavior are shaped strongly by cultural environments rather than by fixed biological instincts.[20]

Ethnographic studies have documented wide variation in domains such as kinship systems, social organization, moral reasoning, and communication practices. If many

[20] This is one way in which my view of culture differs from that of Chagnon (1968), since what he attributes to nature, I largely attribute to nurture, based on my field research on nearby Amazonian communities that behave very differently from the Yanomami, yet inhabiting a very similar ecological niche.

human behaviors were determined primarily by instinctive mechanisms, one might expect greater uniformity across cultures. Instead, anthropologists often find that cultural institutions profoundly shape patterns of behavior (Everett 2012, 2016).

Language provides a particularly striking example. The structural diversity of the world's languages challenges the idea that all languages conform to a single narrowly defined grammatical template (Everett 2005). While some linguistic universals do appear to exist, the range of variation observed across languages suggests that linguistic systems are shaped by cultural evolution and communicative practices as well as by biological constraints.

Anthropological perspectives therefore emphasize the importance of cultural transmission in shaping human behavior. Rather than being determined solely by instinct, many aspects of human cognition emerge through participation in culturally structured environments (Everett 2012; Everett 2016).

5.4 Critiques of Evolutionary Psychology

Another set of criticisms arises from debates within evolutionary psychology. Evolutionary psychologists often explain cognitive traits as adaptations that evolved to solve specific problems faced by ancestral human populations. While this approach has generated valuable insights, critics argue that some evolutionary explanations rely on speculative assumptions about ancestral environments and selective pressures, i.e. they can appear to be little more than "just so" stories.

Philosopher David Buller (2006) has offered one of the most detailed critiques of evolutionary psychology. Buller argues that many claims about evolved psychological instincts are insufficiently supported by empirical evidence. In some cases, the proposed evolutionary explanations are difficult to test because they rely on assumptions about prehistoric environments that cannot be directly verified.

Critics also note that evolutionary explanations sometimes underestimate the role of cultural and developmental processes in shaping cognition. Even if certain cognitive capacities evolved through natural selection, their expression in modern humans may depend heavily on learning and social interaction.

These critiques do not reject evolutionary explanations entirely but emphasize the need for rigorous empirical evidence when proposing evolutionary accounts of cognitive traits. They also highlight the importance of integrating evolutionary perspectives with developmental and cultural approaches.

5.5 Gene–Environment Interaction

Advances in genetics and developmental biology have further complicated traditional notions of instinct. Modern research increasingly demonstrates that genes rarely specify complex behaviors directly. Instead, genes influence developmental processes that interact with environmental inputs at multiple levels.

Gene–environment interactions can occur in many forms. Environmental conditions can influence gene expression through epigenetic mechanisms (Boyd and Richerson 2004), while genetic differences can affect how individuals respond to environmental stimuli. These interactions make it difficult to draw simple distinctions between innate and learned traits.

For example, many studies of language development claim that genetic factors influence aspects of linguistic ability, yet the specific language that individuals acquire depends entirely on the linguistic environment in which they are raised. Similarly, cognitive abilities such as reasoning and problem-solving develop through extended interaction with educational and social environments.

This growing recognition of gene–environment interaction has led many researchers to move away from simplistic instinct-based explanations, especially in relation to human language. Instead, contemporary models emphasize developmental processes in which biological predispositions and environmental influences jointly shape cognitive outcomes (another factor is the language-like abilities of many animals (Everett 2017).

5.6 Implications for Theories of Cognition

Taken together, these critiques suggest that the concept of instinct must be used with caution when explaining complex cognitive phenomena. While biological inheritance undoubtedly influences behavior, the development of cognition typically involves intricate interactions among biological, environmental, and cultural factors.

For the purposes of the present chapter, these critiques are particularly relevant to debates about language acquisition. If linguistic competence were determined entirely by instinctive grammatical structures, one might expect language learning to occur independently of cultural and developmental context. Yet empirical evidence indicates that language acquisition depends heavily on social interaction, communicative experience, and cultural transmission.

These observations do not necessarily refute the possibility that innate cognitive biases influence language acquisition. However, they suggest that such biases operate within broader developmental systems rather than determining linguistic knowledge in isolation.

In this respect, the critiques of instinct discussed in this section help frame the comparison between Peirce and Chomsky developed later in the chapter. Peirce's conception of instinct as a fallible guide to inquiry appears more compatible with contemporary developmental and cultural perspectives than stronger interpretations of instinct as fixed cognitive structure.

Section 6 — Peirce's Theory of Instinct and Abduction

Peirce's TFH was described earlier. It is a formalization of the crucial notion of semeiotic inference in Peirce's distinctive philosophical account of instinct. This section provides a bit more background to better understand TFH. Unlike biological or psychological theories that treat instinct primarily as a fixed behavioral program, Peirce interpreted instinct as a cognitive habit (TFH) guiding the formation of hypotheses about the world. His conception of instinct emerges within his broader theory of scientific reasoning, logica, and semeiotics, particularly his analysis of abductive inference. In Peirce's philosophy, instinct explains how human beings are capable of generating plausible explanatory hypotheses despite the immense complexity of the natural world. This interpretation places instinct at the foundation of scientific inquiry while simultaneously preserving the fallibilist character of knowledge.

6.1 Peirce's Three Types of Inference

Peirce's account of instinct cannot be understood apart from his classification of the three fundamental types of reasoning: deduction, induction, and abduction. These categories represent different logical functions within the process of inquiry.

Deduction involves deriving necessary consequences from general rules. If the premises of a deductive argument are true and the reasoning valid, the conclusion must also be true. Deductive reasoning therefore preserves truth but does not generate new knowledge about the world.

Induction, by contrast, evaluates hypotheses through empirical testing. By examining patterns in observed data, investigators estimate the probability that a hypothesis is correct. Inductive reasoning therefore provides a method for confirming or disconfirming theoretical claims.

Abduction occupies a very different logical role. Abductive reasoning, which Peirce also called "ampliative reasoning" because it can take us to new knowledge, generates explanatory hypotheses that account for surprising observations. Peirce described abduction as the inferential step in which a possible explanation is proposed for an observed phenomenon. In its simplest form, abductive reasoning can be expressed as:

A surprising fact is observed. If hypothesis H were true, the fact would be expected. Therefore, H may be true.

Abduction does not guarantee truth; it merely proposes a plausible explanation that must subsequently be tested. Nevertheless, Peirce regarded abduction as indispensable to scientific discovery because it is the only form of reasoning capable of producing genuinely new ideas.

6.2 The Problem of Hypothesis Formation

Peirce recognized that abductive reasoning raises a profound epistemological problem. Scientific investigation requires investigators to generate hypotheses that explain observed phenomena, yet the space of logically possible hypotheses is virtually infinite. If human beings were forced to search randomly through this vast space of possibilities, the success of scientific inquiry would be extremely unlikely.[21]

Peirce therefore argued that the ability to generate plausible hypotheses must depend on learned habits of the human mind and body and their resonance (not Peirce's term, but along the lines of Gibson 1966; 1979) with their environment. Scientists often produce hypotheses that turn out to be approximately correct even before extensive empirical investigation. According to Peirce, this capacity reflects an evolved affinity between human cognition and the structure of the natural world.

Peirce (CP: 5.173) famously wrote that humans possess "a certain insight, not strong enough to be more often right than wrong, but strong enough not to be overwhelmingly more often wrong than right." This insight is partially what he identified as instinct.

6.3 Instinct as an Evolutionary Adaptation

Peirce's interpretation of instinct was deeply influenced by evolutionary theory. Writing after Darwin's publication of On the Origin of Species, Peirce recognized that human cognitive capacities must themselves be products of evolutionary history. If

[21] Peirce made prominent his understanding of abduction as the formalization of guessing guided by experience, knowledge, or habit (or instinct).

human beings evolved within a world governed by regular natural laws, it is plausible that the human mind developed tendencies that facilitate successful interaction with that world.

Peirce therefore argued that instinctive reasoning reflects an adaptive relationship between mind and nature. Human beings are capable of forming hypotheses that approximate the structure of reality because the human cognitive apparatus evolved within that reality. In this sense, instinct represents an evolutionary bridge between biological adaptation and scientific knowledge.

This perspective differs sharply from rationalist theories that treat knowledge as deriving from innate intellectual principles independent of empirical experience. For Peirce, instinct is neither necessarily a priori knowledge nor purely learned behavior. Instead, it represents an evolutionary predisposition guiding the process of inquiry. Peirce called our instinctive actions "habits," acquired either onto- or phylogenetically (Chomsky's concept of instinct would seem to refer exclusively to phylogentically obtained inference, structures, etc.)

Another important difference between Chomskyan instinct and Peircean inference concerns the evolutionary interpretation of cognitive capacities. Peirce explicitly linked instinctive reasoning to evolutionary adaptation. Because the human mind evolved within the natural world, our cognitive instincts tend to align with the patterns of nature. This evolutionary continuity between mind and world explains why abductive reasoning can produce plausible hypotheses about natural phenomena.

For example, why is it that all humans can hear speech sounds? From an evolutionary perspective, the human auditory system did not evolve in isolation but in tandem with speech production mechanisms and neural specialization. Work by W. Tecumseh Fitch (2000, et. al.) and others shows that changes in the vocal tract (e.g., descended larynx, refined tongue control) expanded the range of possible speech sounds. At the same time, the auditory system became well suited to perceiving those sounds (Fitch 2010). That is, speech systems evolved to match the constraints of human auditory processing. Over evolutionary time, this match-up between ears and vocal apparatus became efficient at decoding and producing the acoustic patterns allowed by human vocal anatomy. Importantly, this matching operates across all Homo sapiens (indeed likely including all Homo species at least since erectus; Everett 2017). That is evolution did not simply provide for English, Pirahã, or Mandarin sounds, but resulted instead in a general capacity for mapping acoustic variation onto linguistic categories**.**

Complementing this perspective, Peirce reasonably treated instinct as an evolutionary bridge between biological adaptation and scientific inquiry. Human reasoning reflects the gradual development of cognitive capacities that enable organisms to navigate complex environments.

Chomsky's interpretation of language evolution is bolder than Peirce's. While he acknowledges that the language faculty must ultimately have biological origins, Chomsky has frequently suggested that the emergence of language may have involved relatively sudden evolutionary developments (Berwick and Chomsky 2017, independent of Darwin's mechanism of natural selection). In some accounts, the key innovation is the computational operation Merge, which enables recursive hierarchical structure. Notice, crucially, here that while Peirce's semeiotic inferentialism, SIGN --> SIGN, can account for *all* knowledge, Berwick and Chomsky's operation of Merge can, by design, only account for the subset of language called syntax which, as we saw earlier, LLMs can learn without Merge.[22] [23] Thus it comes across now as a superfluous proposal from the perspective of LLMs, though LLMs follow the basic semeiotic operation exactly - one sign (number, word, etc) returns or "predicts" another.

This difference highlights contrasting approaches to evolutionary explanation. Peirce emphasized gradual evolutionary continuity, whereas Chomsky has sometimes emphasized the possibility of relatively abrupt cognitive innovations: the former is a gradual theory while the latter is a saltational one.

6.4 The "Natural Light of Reason"

Peirce sometimes described instinctive reasoning using the phrase "the natural light of reason," taken from Galileo's writings (Peirce 1908). Rather than treating this as a source of certain knowledge, Peirce regarded it as a fallible guide to hypothesis formation.

In this respect Peirce's account differs from classical rationalism. Rationalist philosophers such as Descartes and Leibniz argued that certain fundamental truths are accessible through intellectual intuition. Peirce rejected this view, insisting that all human knowledge remains provisional and subject to revision. He defined intuition in the Cartesian sense as "a cognition without any previous cognition" and denied that such cognition was possible.

[22] In other words, if I see or hear one sign, such as a symbol, icon, or index, I respond with another sign - in particular, with a sign that interprets the previous one. This then can be thought of as the basic form of meaning interpretation *and* inference more generally.

[23] This does not mean that LLMs can as of today learn semantics. This is because they do not seem to operate on symbols but on indexes. Therefore, as we saw in the text, Searle's Chinese Room conclusions still hold for LLMs. But I am confident that LLMs will get there one day. Perhaps I should not be so sanguine about the possibilities.

The natural light of reason therefore functions not as a guarantee of truth but as a heuristic mechanism guiding inquiry. Instinct helps investigators identify promising explanatory hypotheses, but these hypotheses must be tested through empirical investigation.

6.5 Instinct and Fallibilism

A central feature of Peirce's philosophy is fallibilism—the doctrine that all human knowledge is potentially subject to error. Fallibilism applies equally to instinctive reasoning. Although instinct helps guide the formation of hypotheses, instinctive judgments can still be mistaken.

Peirce emphasized that the success of scientific inquiry depends on a dynamic interaction between this kind of instinctive conjecture (abduction) and empirical testing. Abductive hypotheses generated through instinct must be subjected to inductive evaluation through observation and experiment.

6.6 Instinct as the Starting Point of Inquiry

For Peirce, instinct-as-abduction represents the starting point of inquiry rather than its conclusion (that is, Peirce would have strongly opposed more recent redefinitions of this term to mean "inference to the best explanation"). Instinctive judgments guide investigators toward possible explanations, but the ultimate authority in science lies with empirical evidence and collective reasoning.

In this way, Peirce's theory integrates biological evolution, cognitive processes, and scientific methodology into a unified account of human reasoning.

Section 7 — Chomskyan Rationalism and the Language Instinct

7.1 Rationalist Intellectual Background

Chomsky's theoretical orientation draws heavily upon the rationalist tradition in philosophy (Chomsky 1968). In several works he explicitly aligns his approach with earlier thinkers such as René Descartes, Gottfried Wilhelm Leibniz, and Wilhelm von Humboldt. These philosophers argued that the human mind possesses inherent structures that shape knowledge and cognition. Rather than deriving knowledge solely from sensory experience, Chomsky argued, the mind contributes organizing principles that make knowledge possible, i.e. TBO.

In Chomsky's interpretation of this intellectual lineage, the rationalist tradition anticipated modern cognitive science by emphasizing the internal structure of the mind. The ability to generate and understand an unbounded number of sentences, for example, suggests that language involves a generative system of rules rather than a collection of learned behavioral patterns. Chomsky therefore interpreted linguistic competence as evidence for the existence of internal mental structures that govern the formation of sentences.

This rationalist orientation marked a significant departure from behaviorist approaches that had dominated American psychology during the mid–twentieth century. Behaviorists generally sought to explain language learning in terms of stimulus–response conditioning and reinforcement. Chomsky argued that such explanations were fundamentally inadequate because they could not account for the creativity and structural complexity of human language.

7.2 The Poverty of stimulus Argument (POS)

Perhaps the most influential argument for instinctive cognitive structure in modern cognitive science arises from linguistics. Chomsky's theory of Universal Grammar proposes that language acquisition depends on innate principles that constrain the structure of possible grammars (e.g. Hauser, Chomsky, and Fitch 2002; Berwick and Chomsky 2017).

The central motivation for this proposal is the POS argument. According to this argument, the linguistic input available to children is insufficient to explain the grammatical knowledge they ultimately acquire. Children rarely receive explicit instruction about grammar, yet they rapidly develop the ability to produce and understand complex sentences.

For example, Chomsky used examples like the following to illustrate the POS:

The man who is tall is here.

**Is the man who tall is here?* vs. *Is the man who is tall here?*

According to Chomsky a plausible hypothesis for forming yes-no questions from assertions might be to move the first verb to the beginning of the sentence. But as the * example (meaning ungrammatical) shows, that doesn't work. What we must do, Chomsky claims, is to move the *main* verb to the beginning. And, Chomsky concludes, children must know the difference between main and subordinate clauses from birth. This is because they apparently never err in this type of question formation.

Let's grant the questionable assumption that they never err. This still is a problematic example because it does not show what it claims to show. And it fails to do this because because it ignores the fact that we ask questions about new information (found in the predicate "is here") not about old information (found in the predicate "is tall" of the relative clause) and that rising intonation beginning at the verb indicates new information. This is not an impoverished stimulus, in other words. The example doesn't work. And this is typical of all such examples. They ignore the multimodality of languages, e.g. gestures, prosody, facial expressions, context, information structures, and the like.

On the other hand, even if such examples fail, if language learning depended solely on inductive generalization from observed sentences, children would face an enormous search problem. The number of logically possible grammatical systems consistent with limited linguistic input is extremely large (but see Cowie 1997). Without additional constraints, identifying the correct grammar would be extremely difficult.

Nativist theories propose constraints to guide the process of language acquisition by eliminating many logically possible but biologically implausible grammars. Their claims are buttressed by acceptance of the POS argument, it remains one of the most prominent arguments for innate cognitive structure in contemporary linguistics (but cf. the discussion of Pullum and Scholz 2002).

It is correct that children typically hear only a limited and often imperfect sample of the sentences that are possible in their language. Just as scientists typically do not have enough information to completely propose a theory before they do or a hunter lacks enough information to tell exactly where their prey has hidden. For all of us in most endeavors, much of the input we receive is fragmented and incomplete. This is the case of children learning their first language. A child hearing an ungrammatical sentence is roughly in the same position as a scientist with a spurious fact. Inference must overcome it. Or as Chomsky would claim, instinct must overcome it. Moreover, children, like scientists, rarely receive explicit instruction about the grammatical rules or generalizations governing their language or incipient theories. Despite these limitations, children reliably acquire the ability to produce and understand grammatically structured sentences by early childhood. Isn't instinct required to account for this?

Chomsky certainly thought so, arguing that this phenomenon cannot be explained solely through inductive learning from experience. If children were required to infer grammatical rules exclusively from the linguistic data available to them, the task would be computationally intractable. The space of logically possible grammars is far too large for such learning to occur without additional constraints. (As this chapter indicates, however, Large Language Models raise serious problems for such claims, by learning most or all syntactic structures without a UG or even meaning, simply by the Peircean FH - SIGN_n --> SIGN_{n+1}.)

The POS argument therefore would seem to suggest that the human mind must contain innate principles that restrict the range of possible grammars. These principles guide the process of language acquisition by limiting the hypotheses that children consider when interpreting linguistic input.

Pullum and Scholz (2002), on the other hand, argue that the classic POS arguments are often stated too vaguely to support strong nativist conclusions. One central problem they identify is lack of explicit formulation: many POS arguments do not clearly specify (i) the exact linguistic knowledge to be explained, (ii) the properties of the input available to the child, or (iii) the precise learning claim being ruled out. Without this rigor, claims that certain constructions are “unlearnable” from input remain impressionistic. Pullum and Scholz emphasize that once corpora and realistic input data are examined more carefully, allegedly absent or rare constructions often turn out to be more available than assumed, undermining the empirical basis of many POS arguments.

A second major issue is insufficient proof of learnability limits. Pullum and Scholz argue that nativist POS arguments typically assert, rather than demonstrate, that no general-purpose learning mechanism could acquire the relevant structures. They point out that alternative models—statistical learning, distributional analysis, and richer interpretations of input (including indirect negative evidence)—are not adequately ruled out. In effect, POS arguments often rely on a false dichotomy between "innate grammar" and "impoverished input," ignoring the possibility that general cognitive learning strategies could exploit subtler regularities in linguistic experience.[24] Again, Piantadosi (2024) seriously undermines POS arguments for syntax, Chomsky's main empirical focus for most of his career, by showing how LLMs learn syntax without UG (and see how TFH actually explains LLMs as part of semeiotic, abductive inference).

According to many critiques, again, the linguistic environment experienced by children may contain far more information about grammatical structure than previously assumed. Large corpora of child-directed speech reveal that caregivers frequently produce highly structured and repetitive linguistic patterns that could provide substantial

[24] As they put it in their abstract: " This chapter examines a type of argument for linguistic nativism that takes the following form: (i) a fact about some natural language is exhibited that allegedly could not be learned from experience without access to a certain kind of (positive) data; (ii) it is claimed that data of the type in question are not found in normal linguistic experience; hence (iii) it is concluded that people cannot be learning the language from mere exposure to language use. We analyze the components of this sort of argument carefully, and examine four exemplars, none of which hold up. We conclude that linguists have some additional work to do if they wish to sustain their claims about having provided support for linguistic nativism, and we offer some reasons for thinking that the relevant kind of future work on this issue is likely to further undermine the linguistic nativist position."

evidence for grammatical learning. The data available to the child must include the multi-modality of language, inference from the cultural and linguistic context, and so on.

Empirical critiques of POS reasoning do not necessarily refute all forms of linguistic nativism. However, they do highlight the need for careful empirical evaluation of claims about the limitations of linguistic input. Understanding the extent to which language acquisition depends on innate structure versus learning remains an open question requiring continued interdisciplinary research.

7.3 The Language Acquisition Device

Chomsky's theory also introduced the concept of a language acquisition device (LAD), a hypothetical cognitive system that enables children to acquire language rapidly and efficiently. Although the term itself appeared primarily in early discussions of generative grammar, the underlying idea remains central to Chomsky's account of language acquisition.[25]

The language acquisition device represents the biological foundation of the language faculty. It includes the innate principles of Universal Grammar as well as mechanisms for interpreting linguistic input. When children encounter sentences in their environment, the language acquisition device uses these principles to construct an internal grammar that generates the sentences of their language.

Importantly, Chomsky's account does not deny the role of experience in language acquisition. Linguistic input is necessary to trigger the development of the grammar and

[25] In Everett 2004, I discuss and reject proposals on the LAD in a prominent book of the time, leading to an interesting exchange with the authors in the Journal of Linguistics, on whether there is a "language organ." That exchange can be found here: https://daneverettbooks.com/wp-content/uploads/2014/04/Language-Organ-review-1.pdf

to determine which parameter settings are appropriate for the language being learned. However, the effects of experience are limited by the constraints imposed by Universal Grammar. But as we have seen, the Chomskyan claims have encountered ever more serious counter proposals.

7.4 The Language Instinct in Popular Discourse

Although Chomsky himself rarely used the phrase “language instinct,” the concept became widely known through Steven Pinker’s 1994 book *The Language Instinct*. Pinker argued that language should be understood as a specialized biological adaptation analogous to other instinctive behaviors found in nature. In Pinker’s interpretation, the human capacity for grammar reflects evolutionary pressures that shaped the cognitive architecture of the brain. Just as spiders possess instinctive abilities to construct webs, humans possess instinctive capacities for language.

While this metaphor captures certain aspects of the generative approach, it has also generated debate. Some critics argue that describing language as an instinct oversimplifies the complex interactions between biological inheritance, cultural transmission, and learning that characterize human linguistic behavior. Peirce would have rejected such a simplistic formulation out of hand.

7.5 Implications for Cognitive Science

Chomsky’s rationalist approach has had a profound influence on cognitive science. By emphasizing the internal structure of the mind, generative linguistics helped shift the study of language away from behaviorist explanations and toward a computational model of cognition.

This perspective contributed to the emergence of the so-called cognitive revolution in psychology and influenced research in fields ranging from artificial intelligence to neuroscience.[26] At the same time, it also sparked ongoing debates about the extent to which human cognition depends upon specialized innate structures.

In the context of the present chapter, Chomsky's conception of the language faculty represents a powerful example of instinct understood as innate cognitive architecture. The contrast between this interpretation and Peirce's account of instinct as a guide to hypothesis formation will become central in the sections that follow.

Section 8 — Language Acquisition Alternatives

8.1 Usage-Based Linguistics and Learning from Input

One of the most prominent alternatives to strong nativist theories of language acquisition is the usage-based framework developed by Michael Tomasello and other researchers in developmental psychology and cognitive linguistics. Usage-based approaches argue that children learn linguistic structures gradually through exposure to patterns of language use in communicative contexts. Rather than relying on an innate grammar containing abstract syntactic principles, children build linguistic knowledge through processes of pattern recognition, analogy, and general cognitive learning.

Tomasello's (2003, 2009) research emphasizes the importance of social interaction in language acquisition. Children learn language within communicative

[26] In Everett in progress, I argue that the "cognitive revolution" should be renamed the "rationalist reboot."

exchanges that involve joint attention, shared intentions, and cooperative interaction with caregivers. All of this presupposes an active role for inference, one that Peirce would have strongly supported. In this context, linguistic constructions are acquired as meaningful units associated with particular communicative functions.

For example, children may initially learn concrete constructions such as “I want X” or “Give me X” as whole patterns rather than as abstract syntactic rules. Over time, they gradually generalize these patterns into more flexible constructions that can be used in novel contexts. This process allows grammatical structure to emerge from repeated usage events rather than from the activation of an innate grammatical blueprint.[27]

Usage-based approaches therefore emphasize the role of domain-general cognitive mechanisms such as categorization, analogy, and statistical learning. According to this view, language acquisition does not require a specialized language faculty containing detailed grammatical knowledge. Instead, linguistic structure emerges from the interaction between general learning abilities and culturally transmitted linguistic input. But all usage-based approaches are simply manifestations of TFH.

8.2 Construction Grammar and the Nature of Linguistic Knowledge

Closely related to usage-based linguistics is the framework known as construction grammar, developed by scholars such as Adele Goldberg (1995), Fillmore, Kay, Michaelis, and Sag (2005), among many others within the cognitive linguistics tradition. Construction grammar proposes that the basic units of linguistic knowledge are

[27] Mitchell Mattox points out to me (pc) that this seems to miss how people improvise in music. But once we have learned the patterns, we have freedom to explore them. In a sense all of human language is improvisation.

constructions, that is, pairings of form and meaning that range from simple lexical items to complex grammatical patterns. This is, without acknowledgement, a Peircean view of grammar.

In this framework, grammatical knowledge consists of a network of constructions (usually symbols in Peirce's sense, but occasionally icons) that speakers learn through exposure to language. These constructions encode both syntactic form and semantic or pragmatic function. For example, the English ditransitive construction ("X gives Y Z") conveys a transfer-of-possession meaning regardless of the specific verbs involved.[28]

Construction grammar challenges the assumption that syntax is governed by abstract rules independent of meaning. Instead, grammatical patterns are understood as learned pairings between linguistic forms and communicative functions, i.e. as a branch of Peircean semeiotics (though Construction Grammar theorists rarely, if ever, acknowledge this clearly). This perspective aligns naturally with usage-based models of language acquisition, since constructions can be acquired gradually through repeated exposure to linguistic patterns.

Empirical research within the construction grammar tradition has demonstrated that children's early linguistic productions often consist of partially learned constructions rather than fully abstract grammatical rules. These findings suggest that grammatical knowledge may emerge incrementally from experience rather than being fully specified in advance by innate grammatical principles.

8.3 Cultural Evolution of Language

[28] Roberts (1997) argues that in Amele there is no need even for a verb in constructions having to do with giving, that the meaning of give is inferred from the construction itself.

Another influential perspective on language acquisition emphasizes the role of cultural evolution in shaping linguistic structure. Researchers such as Everett (2012, 2016), Christiansen and Chater (2022), and Christiansen and Kirby (2003) have argued that languages themselves evolve over generations in response to pressures imposed by human learning and communication.

According to this view, linguistic universals may not reflect innate grammatical constraints but rather the outcome of cultural adaptation. Languages that are difficult for humans to learn tend to change over time as speakers modify linguistic patterns to make them more accessible to learners. Through this process, linguistic structures gradually become better adapted to the cognitive capacities of human learners.

Computational simulations and experimental studies have provided evidence supporting this hypothesis. Iterated learning experiments, in which participants transmit artificial languages across multiple generations of learners, show that languages can develop systematic structure through cultural transmission alone. These results suggest that many features of linguistic structure may arise from cultural evolution rather than from genetically encoded grammatical rules.

The cultural evolution perspective therefore shifts the explanatory focus from innate grammar to the dynamic interaction between learning mechanisms and culturally transmitted linguistic systems. In this framework, the structure of language reflects the cumulative effects of many generations of cultural adaptation.

8.4 Anthropological Perspectives and Linguistic Diversity

Anthropological research has also contributed significantly to debates about language acquisition and linguistic universals (e.g. Schieffelen and Ochs 2012). Studies of languages from diverse cultural contexts reveal a remarkable range of structural variation that challenges the assumption that all languages conform to a narrow set of universal grammatical principles.

Field research conducted among small-scale societies has been particularly important in this regard. Investigations of languages spoken in Amazonia, New Guinea, and other regions have documented grammatical patterns that differ substantially from those found in well-studied Indo-European languages. For example, research on the Amazonian language Pirahã (Everett 2005) has been interpreted by some scholars as challenging the claim that recursive syntactic structures are universal features of human language. Such cases highlight the importance of empirical research on linguistic diversity in evaluating theoretical claims about universal grammar.

Anthropological approaches emphasize that language acquisition occurs within rich cultural environments in which linguistic practices are embedded within social interaction, economic activity, and shared cultural knowledge. This perspective underscores the role of culture in shaping linguistic systems and the processes through which they are learned.

8.5 Implications for Theories of Instinct

The diversity of perspectives surveyed in this section illustrates the complexity of debates surrounding language acquisition. While nativist theories emphasize the role of innate grammatical structures, alternative frameworks emphasize learning, cultural transmission, and social interaction.

From the perspective of the present chapter, these debates are particularly significant because they reveal different interpretations of what instinct might mean in the context of language. In Chomskyan linguistics, instinct corresponds to detailed innate grammatical structure. In usage-based and cultural evolution approaches, by contrast, instinct may be understood more modestly as a set of general cognitive biases that shape learning processes.

This distinction echoes the contrast between Chomsky's rationalist conception of instinct and Peirce's pragmatist interpretation of instinct as a guide to hypothesis formation. The next section will examine this contrast directly by comparing Peirce's theory of instinct with the nativist framework of generative linguistics.

Section 9 — Peirce vs. Chomsky: Two Conceptions of Instinct

9.1 Instinct as Hypothesis Formation vs. Instinct as Cognitive Structure

As seen earlier, the most fundamental difference between Peirce and Chomsky concerns the nature of instinct itself. In Peirce's philosophy, instinct is best understood as an inferential procedure, the basic operation given above, guiding the formation of explanatory hypotheses. Human beings (and all living creatures) possess evolved cognitive dispositions for the basic operation that allow them to generate plausible conjectures about the structure of the world. These conjectures arise through abductive reasoning and serve as starting points for further investigation.

Peirce did not treat instinct as a repository of explicit knowledge. Instead, instinct shapes the direction in which investigators search for explanations but does not determine the final outcome of inquiry. Hypotheses generated through inference remain provisional and must be tested through empirical observation and experimental verification. According to Peirce, inference based on interaction leads to grammatical form and linguistic learning generally.

9.2 The Role of Learning and Experience

Peirce regarded learning as a social enterprise central to the growth of knowledge. Inference initiates inquiry by suggesting or testing hypotheses, but knowledge advances through the interaction between abductive conjecture, deductive reasoning, and inductive testing in social interaction or dialogue. Inference over experiments therefore plays a critical role in determining which hypotheses survive.

Because any reasoning, instinctive, inferential or otherwise, is fallible, empirical investigation is necessary to correct errors and refine theoretical understanding. Knowledge emerges through a collective process of inquiry in which investigators evaluate evidence and revise their beliefs accordingly. Truth is what all inquirers agree on at the "end of inquiry," according to Peirce, a concept that also applies to language learning.

9.3 Epistemological Foundations: Fallibilism vs. Rationalism

The contrast between Peirce and Chomsky also reflects deeper epistemological differences, an extension of their disagreements (in principle, not in historical fact) about language-learning. Peirce's philosophy is grounded in fallibilism—the principle that all human knowledge is provisional and subject to revision. Scientific inquiry progresses through a continuous process of hypothesis generation, testing, and refinement.

Because instinctive judgments can be mistaken, Peirce emphasized the importance of empirical verification and communal inquiry. Knowledge does not originate from infallible intellectual intuition but from the self-correcting processes of scientific investigation. And this same process is crucial to language acquisition. Semeiotic inference produces language, when combined with an open-ended symbol system.

9.5 Implications for Linguistic Theory

The contrast between Peirce and Chomsky has significant implications for cognitive theory. If instinct just is guide to hypothesis-formation, as Peirce suggested, then linguistic knowledge may emerge from general cognitive processes interacting with cultural and communicative environments. Language acquisition would involve exploratory reasoning shaped by both biological predispositions and experiential learning.

In contrast, if instinct consists of detailed innate grammatical structure, as proposed in generative linguistics, then linguistic theory must focus on identifying the formal properties of Universal Grammar. The primary task becomes discovering the structural principles encoded within the language faculty.

These alternative perspectives correspond closely to contemporary debates between generative linguistics and usage-based or construction-based approaches. Usage-based frameworks emphasize learning, cultural transmission, and general cognitive mechanisms, while generative approaches emphasize specialized innate grammatical structure.

9.6 Toward a Comparative Perspective

Although Peirce and Chomsky developed their theories in very different intellectual contexts, comparing their views provides valuable insight into the broader question of instinct in human cognition. Both thinkers recognize that human intellectual capacities cannot be explained solely through environmental learning. Yet they propose different models of how innate tendencies (habits) operate within cognitive systems.

Peirce's theory highlights the importance of inferential reasoning and the evolutionary continuity between mind and nature. Chomsky's theory highlights the structural organization of the cognitive system and the possibility of domain-specific mental faculties.

Understanding the relationship between these perspectives remains an important task for contemporary research in linguistics, philosophy, and cognitive science. By examining how inference vs. instincts compare in both hypothesis formation and linguistic competence, scholars may develop more integrated accounts of human cognition.

10. Toward a Clearer Understanding of the Biological Foundation

This contrast suggests that debates over instinct are often distorted by an overly narrow set of alternatives, i.e. failing to recognize TFH and the contrast between TFH vs.

TBO for general theories of cognition and biology. If instinct is understood only as either rigid innate knowledge or as a discredited relic of pre-scientific psychology, then the conceptual terrain becomes needlessly polarized. A more productive framework would distinguish strong nativist claims from weaker but still biologically serious claims about bias, preparedness, and inferential direction. The question is not whether biology matters. It plainly does. There are no languages invented and used by humans that do not entail biology. The question must be "What contributions precisely does biology make to language-learning and cognition more generally?" For example does it contribute the rich structures and instincts associated with Chomsky's research program? Or does it "simply" support the inferential reasoning process, TFH, of Peircean inference?

A useful starting point is the broader biological background. In evolutionary theory, inherited tendencies are real, but they need not take the form of complete behavioral routines. Organisms inherit developmental dispositions, sensory sensitivities, action repertoires, timing constraints, and species-typical ways of responding to recurrent features of the environment. Classical ethology emphasized patterned behavior that emerges reliably without extensive instruction, and Darwinian theory provided the historical logic for explaining why such patterns exist at all (Darwin 1859; Tinbergen 1951; Lorenz 1965). Yet even in biology, the most defensible accounts of instinct no longer treat genes as encoding finished behaviors in any simple sense. Development is mediated, plastic, and interactive. What evolution supplies are not complete performances but structured propensities realized in relation to ecological and developmental conditions.

If that is right in biology, then it is plausible that the same lesson should carry over to cognition and language and communication. Human beings are not blank slates. Not even Locke or Aristotle would have claimed that in its extreme form. But neither are we necessarily born with fully articulated linguistic or conceptual structures prior to experience. What we inherit are capacities, constraints, sensitivities, and tendencies that make some forms of learning easier than others and some hypotheses more natural than others.

This is where Peirce becomes especially valuable. His account of inference addresses a central problem in epistemology and cognitive development alike: how do agents arrive at promising hypotheses in the first place? Deduction can unfold the consequences of what is already assumed; induction can test and correct what has been proposed; but neither by itself explains the origin of explanatory conjecture. Peirce's answer is that inquiry begins from a biologically, culturally, and historically formed capacity to guess with some limited success. Human beings do not generate explanations at random. They are oriented, however imperfectly, toward certain possibilities rather than others. Peirce called this orientation instinct, though his use of the term differs markedly from theories that identify instinct with fixed behavior or explicit innate knowledge (cf. Everett 2024; in progress, for more details).

The importance of this Peircean insight is that it preserves both biological realism and epistemic fallibilism. Neither instinct nor inference is infallible. Neither guarantees truth, nor does they exempt inquiry from empirical testing. But inference, as opposed to an innate grammar, helps explain why inquiry can begin at all either in language acquisition or in any other endeavor of human inquiry. If organisms had no evolved attunement to the world they inhabit, their search through possible explanations would be hopelessly unconstrained. That they succeed as often as they do suggests some degree of fit between mind and world, but that fit is probabilistic, not infallible. Peirce's formulation thus avoids two extremes at once: the idea that knowledge is simply read off from experience and the idea that it is largely preloaded in cognitive structure.

Notice too, crucially, that TFH is not only not limited to language, but it is not limited to any single modality or level of linguistic analysis. It can handle the combination, say, of words and intonation and gestures, and discourse as well as sentences and conversations. And TFH can draw on all cultural and historical knowledge, depending on the speaker's experience, and thus obviates much of the reasoning behind the poverty of the stimulus arguments, as we have discussed earlier. The basic operation underlies *all* language.

This way of thinking also aligns more naturally with several strands of contemporary developmental research. Studies in infant cognition, for example, show that children begin life with non-trivial capacities for tracking objects, distinguishing quantities, or responding to social cues (Spelke 2000; Carey 2009). These findings are important, but they do not force the conclusion that infants possess mature conceptual systems in miniature. It is more plausible to say that they begin with predispositions that guide subsequent learning (perhaps along the lines of Solomonoff 1964, responding to Putnam 1963 on the problematic nature of "priors" in Bayesian reasoning).

The same general lesson applies to language. Chomskyan linguistics was surely correct to insist that language acquisition cannot be understood as mere imitation or passive accumulation of strings. Children acquire language rapidly, creatively, and under imperfect input conditions. Any adequate theory must explain that fact. But it does not follow that the only explanation is a richly specified Universal Grammar containing detailed formal constraints on all possible human languages. The failure of simple empiricism does not automatically vindicate strong nativism. Pragmatism unifies the best of both. How can children learn language? By inference. And why can every person learn a language? It is not because language is rigidly fixed in structure and inventory - there is too much linguistic variation in the world for that.[29]

Alternative approaches in linguistics and cognitive science suggest a more graduated view. Usage-based theories argue that children learn from structured input in communicative contexts, drawing on pattern recognition, analogy, intention-reading, joint attention, and statistical sensitivity (Tomasello 2003). Construction grammar similarly treats linguistic knowledge as a repertoire of learned form-meaning pairings ranging from local patterns to abstract schemas (Goldberg 1995). These approaches do not deny biology. But they do attribute to it a different role than that envisaged in the Chomskyan paradigm.

In addition, research on cultural evolution has altered the direction of explanation. Instead of assuming that universal features of language must come directly from an innate grammar, one can ask how languages themselves adapt to learners across generations. On this view, linguistic systems are shaped by repeated cycles of learning, use, and transmission. Structures that are easier to process, acquire, and reproduce tend to

[29] If languages were rigidly given by UG in Chomsky's sense, i.e. on the genes, then we would expect evolutionary pressures to make some languages unlearnable for some populations as genotype varies via local niche pressures. Lactase permanence, for example, is only a few thousand years old in some populations, roughly the same age as the "parameter" of pro-drop in Chomskyan theory (i.e. that some languages can drop their subjects, as in Portuguese, while others, e.g. English and French, cannot).

persist, while less stable structures tend to erode or be reorganized (Kirby 2002; Christiansen and Chater 2016). This does not eliminate biological constraint, but it distributes explanation across both biological and cultural timescales. Human learners shape language, and language evolves in response to those learners (see especially Deacon 1997).

Taken together, these lines of work suggest that the most fruitful alternative framework (e.g. in Everett 2016) is a model of cognition that, in effect, provides for what we might call "constrained plasticity." Organisms inherit biases, sensitivities, and inferential tendencies that make certain developmental trajectories more likely than others. These constraints matter greatly, but they do not dictate a single fixed outcome. Development remains ontologically situated, environmentally dependent, and culturally mediated.

Under such a framework, instinct would be best understood as probabilistic developmental orientation, along the lines of TFH, where the sign to the right of the arrow is constantly adjusted by tacit or explicit knowledge. It would name those inherited features of organisms that channel attention, pattern detection, hypothesis formation, and learning without themselves amounting to finished knowledge. Some instincts may be perceptual, making certain distinctions more salient than others. Some may be social, disposing organisms toward cooperation, imitation, or shared attention. Some may be inferential, biasing inquirers toward particular explanatory forms. In language, such instincts might include sensitivity to communicative intention, sequential patterning, hierarchical grouping, vocal discrimination, or symbolic conventionalization. These are outgrowths of TFH.

This conception has several advantages. First, it avoids the conceptual rigidity associated with older instinct theories. If instinct is treated as a flexible bias rather than a fixed program, one can acknowledge biological preparedness without denying developmental variability. Second, it fits better with contemporary biology, which increasingly emphasizes gene-environment interaction, developmental systems, and the layered organization of traits. Third, it allows one to preserve what was most compelling in the Chomskyan challenge to behaviorism—the insistence that learning is nonrandom and partially driven by biology —without committing to the strongest claims of formal nativism. Finally, it resonates with Peirce's broader philosophy, in which cognition emerges through an ongoing relation between the organism, the world, and inquiry.

Peirce's TFH also clarifies the role of fallibility. If instinct is a biasing system, not a guarantee, then error is not an anomaly but an expected feature of cognition. Organisms are guided, not omniscient. Their inherited tendencies can lead them efficiently toward useful generalizations, but also toward oversimplification, illusion, or local failure. This is precisely why learning, correction, and culture matter. A theory of cognition should therefore explain not only why humans succeed in acquiring knowledge, but also why that success depends on iterative interaction with environments and communities of practice.

The broader philosophical payoff is considerable. A weaker, bias-based conception of instinct makes it possible to integrate biology and culture, respecting the significant contributions of each. It also allows inquiry itself to be treated as an evolved achievement. Peirce's great insight was that the human mind is not merely a passive receptor of impressions nor a warehouse of preformed truths. It is an adaptive, fallible, hypothesis-forming system. Chomsky's great insight, by contrast, was that cognition is constrained and cannot be explained by unstructured association alone.

The most promising conclusion, then, is not that instinct with regard to language should be abandoned, but that it should be redefined. The concept remains useful when it refers to inherited orientations that shape what organisms notice, how they learn, and which possibilities they are disposed to explore.

Language learning in the sense presented here is the outcome of a biologically prepared but developmentally open system interacting with historically evolved linguistic practices and culminating in TFH or TBO contributes to language learning, meaning, the evolution of language, and other manifestations of communication, but as guidance rather than blueprint. Such a model is more consistent with modern biology, more flexible in relation to linguistic diversity, and more faithful to the fallibilist insight that cognition develops through engagement with a world that is never simply given and never wholly pre-scripted in the mind.

Section 11 — Conclusion

The concept of instinct has occupied an important yet controversial position in the study of human cognition and behavior. Throughout the history of philosophy, psychology, and biology, scholars have invoked instinct to explain patterns of behavior that appear early in development, occur widely across individuals, and operate without explicit instruction. At the same time, the concept has often been criticized for its conceptual ambiguity and for its tendency to function as a placeholder explanation when underlying mechanisms remain poorly understood. Contemporary debates about language acquisition illustrate the continuing relevance of these tensions.

The primary contribution of this chapter lies in the comparison between Charles Sanders Peirce and Noam Chomsky, via Chomsky's TBO (Merge) and Peirce's TFH, SIGN --> SIGN. These thinkers represent two distinct ways of understanding instinct within theories of cognition. In Peirce’s pragmatist philosophy, instinct operates as an evolutionary guide to abductive reasoning. Human beings possess tendencies that direct attention toward plausible explanatory hypotheses, but these hypotheses must be tested through empirical inquiry. Knowledge therefore develops through a fallible and self-correcting process.

By examining contemporary debates in language acquisition, the chapter has argued for the greater generality and accuracy of TFH. Evidence from usage-based linguistics, construction grammar, and cultural evolution research suggests that linguistic structure emerges through dynamic interactions between learning processes and cultural transmission. Within this context, Peirce’s conception of instinct as a probabilistic guide to inquiry provides a promising framework for integrating biological predispositions with developmental and cultural processes.

Understanding human cognition therefore requires moving beyond the simple opposition between instinct and learning. Cognitive development reflects the interplay of evolutionary history, inferential reasoning, social interaction, and cultural evolution. Future research in linguistics, philosophy, psychology, and anthropology will continue to refine our understanding of how these factors interact to produce the remarkable flexibility and creativity of human thought.

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